

Graduate Research Plan Statement

Near-infrared imaging is vital for identifying and analyzing stars and galaxies at high redshifts. These photons probe ultraviolet (UV) light from high redshift galaxies, where UV colors are sensitive to star formation, dust, and metallicity. During the Epoch of Reionization (EoR), galaxy stellar populations and dust enrichment rapidly develop compared to average timescales we observe in our local universe, but there is still no single conclusion as to how this evolution occurred. With the launch of the *James Webb Space Telescope (JWST)* and its ability to probe the near-infrared in significantly greater detail than ever before, we can peer further into the early universe and obtain detailed constraints on galaxy evolution at the earliest times.

With access to early reductions of Cycle 1 program data with *JWST*, I will analyze the rest-frame UV spectral slope, β , and its correlation with stellar populations at redshifts $z \sim 8-14$. At these early epochs, it is predicted that stars with very blue, ultra-low metallicities may have resided within galaxies. These ultra-low metallicity galaxies ($\beta \lesssim -3.0$) could contain Population III stars with metallicity $Z_* \lesssim 10^{-3} Z_{\odot}$ (Schaerer et al. 2003), or very young Population II stars enriched by only Population III-produced metals (Jaacks et al. 2018)! Typical evolutionary timescales for dust enrichment are $10^8 - 10^9$ years; however, at $z > 5$ ($\sim 10^9$ years after the Big Bang), dust enrichment is already easily observable (Morgan & Edmunds 2003). Many have postulated this to be due to spontaneous grain growth occurring in such a short time (a few million years long, Mattsson et al. 2015), but factors that caused this rapid dust formation are still heavily debated. Thus, by analyzing the evolution of the UV spectral slope, I will be able to refine any implications it may have on dust growth due to the stellar populations that existed during the EoR and the attenuation it caused.

Intellectual Merit

With advancements in near-infrared (NIR) imaging with the *Hubble Space Telescope (HST)*, and more recently with the *JWST*, observers are given access to rest-frame UV emission from early galaxies, which is of key importance when trying to understand galaxy physical properties at high redshift. Although traditionally measured via spectroscopy, this quantity can also be measured photometrically if a galaxy has detections in at least two photometric bands in the rest-frame UV.

To prepare for my proposed project, I am currently studying the UV spectral slope at $z \sim 2-4$. While measurements of the UV spectral slope for typical ($\lesssim L^*$) galaxies were previously unavailable at $z=2-4$ due to the lack of Lyman-break coverage with the CANDELS survey (Grogin et al. 2011), once space-based UV ($0.275 \mu\text{m}$) imaging with UVCANDELS (Teplitz et al. 2018) was obtained, data was available to select galaxies via the Lyman-break. I am currently using UVCANDELS + CANDELS imaging data to measure β at redshifts $z=2-4$. Here, I measure individual photometric redshifts and implement spectral energy distribution (SED) fitting which results in the best-fit synthetic stellar population model. I can measure the UV spectral slope directly from these best-fit spectra using power-law fitting (following Finkelstein et al. 2012). **With this work, I will equip myself with the necessary skills and techniques to extend these studies to higher redshifts with *JWST*.**

At $z \geq 9$, there are very few estimates for what the UV spectral slope and corresponding galaxy properties look like (e.g., Tacchella et al. 2021). By studying the stellar populations of these most distant galaxies with advanced *JWST* imaging, I can explore changes in UV colors in greater detail, and its dependence on stellar mass, age, metallicity and dust extinction. **With current access to expert-led reductions with *JWST* early in Cycle 1, I will be amongst the first to measure β out to $z \sim 10-14$ and anticipate 4 key papers to come from these studies.**

Phase 1: The *JWST* CEERS Survey (Finkelstein et al. 2017) and NGDEEP Survey (Finkelstein et al. 2021) are amongst the first surveys with data collection complete by early 2023. My PI (co)-leads both teams; thus, I have early access to reduced data. I will first implement the same techniques with my $z=2-4$ samples to sources at $z \sim 9-13$ with CEERS, then extend to the highest redshifts, $z \gtrsim 14$, with the deeper NGDEEP Survey. **CEERS NIRCам will detect ~ 25 newly discovered $z \sim 9-13$ modestly-massive galaxies.** With six filters in all 10 of NIRCам's pointings, CEERS achieves imaging at depths of 5σ at $m_{\text{AB}} = 28.6$, **nearly ~ 2 mag deeper than CANDELS in this field** – but at redder wavelengths. NIRCам

imaging will allow us to obtain rest-frame UV spectral slope measurements made with 2-3 rest-UV colors for galaxies at $z=10-14$ for the first time. **This advancement in photometry pushes this technique to higher redshifts than ever before and allows us to implement our method of SED-fitting with multiple rest-UV colors.**

Phase 2: The NGDEEP Survey is a deep spectroscopic and imaging survey targeting constraints in feedback processes in galaxies across cosmic time. NGDEEP's parallel ultra-deep NIRCам imaging in the Hubble Ultra-Deep Field (HUDF)-Par2 field will detect $\sim 25-75$ galaxies at redshifts $z=10-14$ over 5 arcmin^2 at $1 - 5 \mu\text{m}$ on the deepest *HST* F814W imaging. NGDEEP reaches $m \sim 30.6$, up to ~ 0.5 mag deeper than Guaranteed Time Observation NIRCам surveys. **I will provide robust measurements for the UV spectral slope and galaxy morphology, namely for UV-faint, low-mass galaxies that may go undetected with wider survey imaging and are perhaps home to ultra-low metallicity stars.**

Phase 3: After completing my first look into high redshift galaxies with my first two *JWST* projects, I will revisit samples previously measured at $z=4-8$ by *HST* (Finkelstein et al. 2012) with > 4 rest-UV colors with CEERS NIRCам and obtain data across 5 *JWST* programs (that will be public access by this phase): CEERS, NGDEEP, PRIMER (PI: J. Dunlop), JADES (Co-PIs: M. Rieke + P. Ferruit), and COSMOS-Web (Co-PIs: J. Kartaltepe + C. Casey). **With these measurements, I will develop the most comprehensive investigation into the evolution of the UV spectral slope at $z=2-14$.** My timeline will focus on the composition of stars and dust in galaxies, where I will map out how galaxies became chemically enriched across cosmic time, possibly constrain dust timescales, and discover ultra-low metallicity galaxies. **Successful retrieval of the evolution of these properties will serve as proof of concept and set the stage for the final phase of this proposal.**

Phase 4: This component includes a collaboration with Dr. Danielle Berg at the University of Texas at Austin. *JWST* NIRSpec archival data from Cycle 1 programs will span each CANDELS field and the sources for which I measure β . Here, I will implement a detailed follow-up spectroscopic analysis for every galaxy in my sample and compare them with sources in the $z \sim 0$ *HST* CLASSY catalog (Berg et al. 2022), which targets low redshift galaxies with similar properties to $z \gtrsim 2$ galaxies, such as stellar mass, star formation rate, and reddening. These samples act as a training set for star-forming galaxies across all redshifts and will help me interpret any relationships between β and galaxy properties at very high redshift, such as gas-phase metallicity, the ionization parameter, and dust attenuation. With this, we can distinguish physical reasons behind differences in β and determine if the same is true for their $z \sim 0$ galaxies.

Broader Impacts

As someone who has significantly benefited from outreach, I aim to demonstrate my research findings to the public and mentor minorities and women in STEM as they advance in their undergraduate careers and beyond. Dr. Steven Finkelstein leads UT Austin's Astronomy Vertically Integrated Projects program, where undergraduates work with each other and Dr. Finkelstein's research group on both *HETDEX* and *JWST*-related projects, intending to improve STEM retention rates. Being a mentor in this group has allowed me to mentor undergraduates, offer them research experience, and help solidify their goals and decisions by the time they leave the program. I plan to continue outreach activities similar to the ones I have led and experienced as an undergraduate (as described in my personal statement). I would also like to use these outreach opportunities to educate the public on my research and the progress in any findings that are made possible with access to *HST* and *JWST*. In addition to educating the public, I am part of the DEI organization in my department that revolves around inclusivity and equality amongst underrepresented faculty, post-doctoral and graduate students in hopes of making the department a safe and equitable work environment. From experience, opportunities to reach out to minority groups in STEM have been highly beneficial both on the giving and receiving end. And as I continue on throughout my career, I hope to make purposeful contributions to DEI efforts, the understanding of our Universe, and offer guidance to those in need.

References Berg et al. 2022, Brammer et al. 2008, Calzetti et al. 1994, Carnall et al. 2018, Finkelstein et al. 2012, 2017, 2021, Grogin et al. 2011, Jaacks et al. 2018, Mattsson et al. 2015, Morgan & Edmunds 2003, Schaerer et al. 2003, Teplitz et al. 2018, Tacchella et al. 2021